

Education

Charles University & Saarland University

Prague, CR | Saarbrücken, DE

EUROPEAN MASTER'S PROGRAMME IN LANGUAGE & COMMUNICATION TECHNOLOGIES

2021 - 2025 (study break b/w

12/23-03/25)

- **Thesis:** Does multilingual training improve the performance for low-resource language summarization?
- **Projects:**
 - Summarization System for context retention in generating theatre script with customized GPT-2 beyond 1024 tokens for THEaiTRE project
 - BART-based summarization system for processing Shakespeare's plays in English and German

Bhagalpur College of Engineering

Bhagalpur, IN

B.TECH IN COMPUTER SCIENCE ENGINEERING

2014 - 2018

- **Major Project:** e-Dustbin: A fully automated dustbin using Arduino to optimize route for waste pickup vehicles
- **Minor Project:** Finding weakest node in a network for attack and maximum damage using graph theory

Publications

GrEmLIn: A Repository of Green Baseline Embeddings for 87 Low-Resource Languages Injected with Multilingual Graph Knowledge

NAACL'25 Findings

Daniil Gurgurov, **Rishu Kumar**, Simon Ostermann

UM-DFKI Maltese Speech Translation systems

IWSLT'24

Kumar Rishu, Aiden Williams, Claudia Borg, Simon Ostermann

FINDINGS OF THE IWSLT 2023 EVALUATION CAMPAIGN

IWSLT'23

Multiple authors

UM-DFKI Maltese Speech Translation

IWSLT'23

Aiden Williams, Kurt Abela, **Rishu Kumar**, Martin Bär, Hannah Billingham, Kurt Micallef, Ahnaf Mozib Samin, Andrea DeMarco, Lonneke van der Plas, Claudia Borg

Team ÚFAL @ CreativeSumm: BART and SamSum based few-shot approach for creative Summarization

CreativeSumm'22 (COLING Workshop)

Rishu Kumar, Rudolf Rosa

Short-Term Word-Learning in a Dynamically Changing Environment

arxiv

Christian Huber, **Rishu Kumar**, Ondřej Bojar, Alexander Waibel

Team ÚFAL at CMCL 2022 Shared Task: Figuring out the correct recipe for predicting Eye-Tracking features using Pretrained Language Models

CMCL (ACL 2022)

Sunit Bhattacharya, **Rishu Kumar**, Ondřej Bojar

ELITR multilingual live subtitling: Demo and strategy

EACL 2021

Multiple Authors

Operating a Complex SLT System with Speakers and Human Interpreters

MTSummit 2021

Ondřej Bojar, Vojtěch Srdečný, **Rishu Kumar**, Otakar Smrž, Felix Schneider, Barry Haddow, Phil Williams, Chiara Canton

SEntFiN 1.0: Entity-Aware Sentiment Analysis for Financial News

JASIST

Ankur Sinha, Satishwar Kedas, **Rishu Kumar**, Pekka Malo

Fine-tuning, Retrieval, or Merging? Augmenting LLMs with Search Task Knowledge

in progress

Main Contributor

Work Experience

IRIT

Information retrieval and
Information synthesis

INTERN

Apr - Aug 2025

- Exploring the performance difference of LLMs with a knowledge base in comparison to LoRA fine-tuning
- Evaluating the performance between random context for inference time learning w.r.t topic-augmented context
- Working on model merging library to select the best merging strategy and parameters
- Participation in TREC-iKAT for intent classification and response generation

Legaleey Inc.

RESEARCH CONSULTANT

July 2024

- Researched LLM-based parsing of PDFs and pdftotext + heuristics for knowledge base creation
- Incorporated parsing pipelines for clean and semantically relevant output in Elastic Search
- Expanded legal data extraction points to create a minimal viable product based on CUAD
- Moved the tech stack from Generative models to extractive models for efficient and trustworthy results

DFKI

Efficient and Explainable NLP Group
(MLT)

RESEARCH ASSISTANT / JR. RESEARCHER

Nov 2022 - June 2024

- Worked on creating intentionally erroneous data to check automated evaluation for summarization
- Co-organized Dialectical and Low-Resource SLT shared task in IWSLT ('23, '24)
- Created baseline system for evaluation and ran evaluation campaign
- Participation in IWSLT shared tasks for low-resource languages ('23, '24)

Saarland University

Language Science & Technology

TUTOR

Oct 2022 - Feb 2024

- Tutor for Neural Networks Theory and Implementations in WS'22 & Statistical NLP in SS'23
- Proposed and trialled a course, "Basic tools for NLP," which covers bash, git, latex, etc., in SS'23; Confirmed as a regular course in WS'23

European Live Translator (Charles University)

Institute of Formal and Applied
Linguistics

RESEARCH INTERN

Aug 2020 - Mar 2022

- Prepared pipeline for continuous evaluation of ELITR-testset
- Improved SLTev a tool to evaluate ASR, MT and SLT performance
- Domain adaptation/retraining of ASR models (English and Czech) to improve quality and subsequently produce a better translation
- Maintained and improved bash pipelines to connect ASR, MT, and publishing workers
- Stress testing of systems to ensure reliability before using in research demo at multiple venues
- Created pipelines for automation of real-time meeting summarization of meeting in collaboration with alfaview and PerVoice

Indian Institute of Management Ahmedabad

Production & Quantitative Methods

RESEARCH ASSISTANT

Mar. 2018 - Jul. 2020

- Developed lexicon-based language models capable of extracting entity-based sentiments; Average news headline length ~ 8 words
- Released an annotated dataset of 12,000+ financial news headlines
- Compared GBMs, SVMs, and MLPs for classification; Contrasted feature-engineered input vectors vs GloVe and Word2Vec; Achieved ~ 91% accuracy and ~ 93% F1-score using Roberta
- Validated the effect of sentiment information contained in news flow on the over-night aggregate market (NSE 500 index) returns
- Audited a PhD level course on Non-Linear Optimization

Academic Misc

- Program Committee member for IWSLT'23, '24, '25
- Student Council Member for Language Science & Technology programme for 2022-23
- Emergency reviewer for Workshop on Information Extraction from Scientific Publications @ ACL-IJCNLP 2022
- Erasmus+ scholarship (20210-23)
- Student volunteer at COLING'22
- Student Representative for European Master's Programme in Language Science and Technologies for 2021-22
- Secured a rank of 3,978 in Graduate Aptitude Test in Engineering'19 among 100,000+ candidates
- Started a coding club and organized events to promote pair-programming
- Attended an Undergraduate Summer School at Computer Science and Automation Dept., Indian Institute of Science, Bangalore
- All India Rank 419 in IndiaHacks'16, an algorithm challenge on HackerEarth; College Rank 1

Technical Knowledge

- Programming** Python, C, C++, bash, Matlab, HTML, TS, Rust
- Frameworks** PyTorch, TensorFlow, Keras, Scikit-learn, MySQL, Marian
- Miscellaneous** LaTeX, SLURM, SGE, *nix

Languages

Fluency	Native	C2	B1	A2	A1
Languages	Hindi, Angika	English	Gujarati, Bengali, Bhojpuri, Maithili	German	Czech, French